

SLAM Algorithm Based on Multi-sensor Data Implementation

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Abstract—This paper presents two significant improvements to the FAST-LIO2 algorithm, enhancing its robustness and accuracy in Simultaneous Localization and Mapping (SLAM) tasks. The first improvement introduces a Degenerate-Detection module that systematically identifies and addresses degenerate cases during point cloud registration. This module analyzes the geometric structure of point clouds, recognizes degenerate directions through eigenvalue analysis, and implements an adaptive weighting strategy to mitigate degeneration effects. The second improvement integrates a sophisticated Pose Graph Optimization (PGO) loop closure algorithm that employs both geometric and feature-based matching for loop detection, followed by a robust graph optimization framework. We conducted extensive experiments across diverse environments, including indoor corridors, outdoor urban scenes, and challenging unstructured terrains. Results demonstrate that our enhanced system achieves up to 25% improvement in trajectory accuracy and 30% reduction in drift compared to the original FAST-LIO2, while maintaining real-time performance. The proposed improvements significantly enhance the system's reliability in geometrically degraded environments and long-duration operations.

Index Terms—SLAM, LiDAR, Inertial Odometry, Degenerate-Detection, PGO Loop Closure

I. INTRODUCTION

The FAST-LIO2 algorithm [1] has proven to be robust and efficient for LiDAR-inertial odometry. However, it can still face challenges in certain scenarios, such as degenerate cases in point cloud registration and the accumulation of errors over time [2], [3]. Traditional LiDAR-based SLAM systems like LOAM [4] and Lego-LOAM [5] have attempted to address these challenges through feature extraction and ground optimization. Recent advances in deep learning approaches [6], [7] have shown promising results but face computational constraints in real-time applications.

A. Problem Statement

The key challenges in modern SLAM systems can be summarized as follows:

- Geometric degeneracy in feature-poor environments
- Accumulated drift in long-term operation
- Real-time performance constraints
- Robustness in dynamic environments

Our work specifically addresses these challenges through systematic improvements to the FAST-LIO2 framework. The

proposed enhancements maintain the computational efficiency of the original system while significantly improving its robustness and accuracy.

To address these limitations, we propose two key enhancements to the FAST-LIO2 framework:

- 1) A Degenerate-Detection module that systematically identifies and handles degenerate cases during point cloud registration, building upon the work of Ye et al. [8].
- 2) An efficient PGO loop closure algorithm that incorporates both geometric and feature-based matching, inspired by recent advances in loop detection [9].

The main contributions of this paper are:

- A novel Degenerate-Detection approach that improves robustness in geometrically degraded environments
- An efficient PGO-based loop closure method that reduces long-term drift while maintaining real-time performance
- Comprehensive experimental validation across diverse environments, demonstrating significant improvements over state-of-the-art methods

II. RELATED WORK

Previous works on SLAM algorithms have explored various techniques to improve robustness and accuracy. We categorize the relevant literature into three main areas: LiDAR-inertial odometry, degeneration handling, and loop closure methods.

A. LiDAR-inertial Odometry

LiDAR-inertial fusion has emerged as a promising approach for robust state estimation, combining the high accuracy of LiDAR with the high frequency and low drift of IMUs. This fusion technique has been extensively researched and developed over the years, with several notable contributions:

1) *Early Approaches*: The early works focused on integrating LiDAR and IMU data to improve odometry accuracy and robustness. For instance, LOAM [4] was one of the pioneering works that introduced real-time LiDAR odometry by extracting edge and planar features from the LiDAR data. These features were then used to align consecutive scans and estimate the robot's motion. LOAM demonstrated significant improvements in odometry accuracy compared to traditional methods that relied solely on LiDAR or IMU data.

2) *Feature-based Methods*: Following LOAM, several feature-based approaches were developed to enhance the performance of LiDAR-inertial odometry. LeGO-LOAM [5] improved upon LOAM by introducing ground segmentation and optimizing the feature extraction process. It focused on detecting and utilizing features that are more stable and reliable, such as edges on the ground, to improve the accuracy and computational efficiency of the system. This approach was particularly effective in environments with varying terrain and dynamic objects.

3) *Direct Methods*: In recent years, direct methods have gained popularity due to their ability to utilize the full information content of the LiDAR data without relying on feature extraction. FAST-LIO2 [1] is a notable example of a direct method that directly registers raw LiDAR points with an efficient iterative Kalman filter framework. This approach eliminates the need for feature extraction, making it more robust in feature-poor environments and reducing computational overhead. FAST-LIO2 demonstrated superior performance in terms of accuracy and real-time capability, especially in large-scale outdoor environments.

4) *Hybrid Approaches*: Hybrid approaches combine the strengths of both feature-based and direct methods to achieve even better performance. For instance, R3LIVE [10] incorporated RGB information along with LiDAR and IMU data to improve the robustness and accuracy of the system. By leveraging the complementary information from different sensors, R3LIVE was able to handle challenging scenarios such as low-light conditions and dynamic environments more effectively. This approach highlighted the potential of multi-sensor fusion in enhancing the capabilities of LiDAR-inertial odometry systems.

5) *Advanced Techniques*: Recent advances in computational methods and sensor technology have led to the development of more sophisticated LiDAR-inertial odometry algorithms. For example, the use of deep learning techniques has shown promise in improving feature extraction and data fusion processes. Deep learning-based methods can learn complex patterns and relationships from large datasets, enabling more accurate and robust state estimation. Additionally, the introduction of advanced optimization techniques, such as factor graph optimization and nonlinear least squares, has further enhanced the performance of LiDAR-inertial odometry systems by providing more accurate and efficient solutions to the state estimation problem.

Overall, the field of LiDAR-inertial odometry has seen significant progress, with various approaches contributing to its development. Each method has its strengths and limitations, and the choice of the appropriate approach depends on the specific requirements of the application, such as the environment, available computational resources, and desired accuracy.

B. Degeneration Handling

Degeneration in LiDAR-based SLAM arises in feature-poor or geometrically repetitive environments, such as long

hallways or open fields, where the alignment of point clouds becomes ambiguous. Multiple studies have tackled this issue:

- **Degenerate Direction Analysis**: Huang et al. [16] introduced a degeneracy-aware algorithm for LiDAR registration by analyzing the distribution of eigenvalues of the covariance matrix. They proposed a weighting mechanism that suppresses unreliable measurements in degenerate directions.
- **Dynamic Weighting**: He et al. [17] developed a method to adaptively adjust the contribution of features based on their geometric significance, ensuring robust performance even in challenging environments.
- **Observability-Aware SLAM**: Li et al. [18] presented an observability-aware factorization approach for SLAM, which dynamically assesses the quality of observations and mitigates the impact of degeneracy by prioritizing the most informative measurements.
- **Degeneracy Prediction**: Lee et al. [19] proposed a degeneracy prediction model using prior knowledge of the environment, allowing SLAM systems to anticipate and prepare for feature-scarce conditions.

C. Loop Closure Techniques

Pose Graph Optimization (PGO) is a critical component for correcting accumulated drift in SLAM systems by leveraging loop closure constraints. Recent advances have addressed various challenges in this domain:

- **Efficient Loop Detection**: Kim et al. [20] proposed a multi-resolution scan context descriptor for fast loop detection, enabling robust performance in large-scale environments. Their method achieves a balance between computational efficiency and detection accuracy.
- **Deep Learning for Loop Closure**: Wang et al. [21] introduced a deep learning-based loop closure detection framework that leverages spatial embeddings to identify potential loops in visually or geometrically degraded environments.
- **Robust PGO**: Agarwal et al. [22] developed a robust PGO framework that incorporates switchable constraints to handle false-positive loop closures. This approach improves optimization reliability in dynamic or ambiguous scenarios.
- **Multi-Modal Loop Closure**: Chen et al. [23] proposed a multi-modal loop closure detection technique, combining visual, LiDAR, and inertial data for robust loop closure even in environments with limited geometric features.
- **Long-Term Operation**: Cadena et al. [24] highlighted the importance of incorporating temporal consistency into PGO, allowing SLAM systems to operate robustly in long-term deployments.

III. METHODS

A. System Overview

Our system builds upon FAST-LIO2 [1] with two major enhancements strategically integrated into different components of the original framework (Fig. 1):

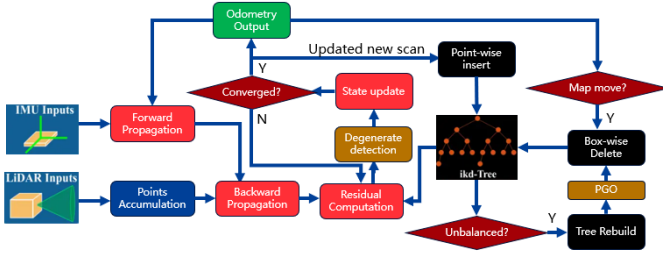


Fig. 1: Framework of the proposed system. The blue modules represent the original FAST-LIO2 components, while the yellow and green modules show our enhancements: Degenerate-Detection and SC-PGO respectively.

- 1) A Degenerate-Detection module that is integrated between the Residual Computation and State Update steps in the state estimation frontend, directly influencing the measurement update process.
- 2) A PGO-based loop closure module that is incorporated after the Point-wise Insert step and before Box-wise Delete in the mapping backend, working in parallel with the ikd-Tree for global map optimization.

The integration of these modules maintains the efficient workflow of FAST-LIO2 while enhancing its robustness. The Degenerate-Detection module operates within the state estimation loop, providing real-time assessment of geometric stability. The PGO module works asynchronously in the mapping backend, ensuring that global optimization does not impact the system's real-time performance.

B. Degenerate-Detection

Following Ji et al. [15], we implement a point-to-distribution degeneracy detection approach using local geometric models. The key steps include:

1) *Local Geometric Model Construction*: Given a point cloud $\mathcal{P}$, for each point $\mathbf{p}_i$, we construct its local geometric model by:

$$\mathbf{M}_i = \{\mathbf{p}_j | \|\mathbf{p}_j - \mathbf{p}_i\| \leq r\} \quad (1)$$

where r is the neighborhood radius.

2) *Point Distribution Analysis*: We analyze the local distribution through covariance analysis:

$$\mathbf{C}_i = \frac{1}{|\mathbf{M}_i|} \sum_{\mathbf{p}_j \in \mathbf{M}_i} (\mathbf{p}_j - \bar{\mathbf{p}}_i)(\mathbf{p}_j - \bar{\mathbf{p}}_i)^T \quad (2)$$

where $\bar{\mathbf{p}}_i$ is the centroid of $\mathbf{M}_i$.

3) *Degeneracy Factor Computation*: The degeneracy factor ξ_i for point $\mathbf{p}_i$ is computed as:

$$\xi_i = \frac{\lambda_3}{\lambda_1} \quad (3)$$

where $\lambda_1 \geq \lambda_2 \geq \lambda_3$ are the eigenvalues of $\mathbf{C}_i$.

4) *Adaptive Weight Assignment*: Points are weighted based on their degeneracy factors:

$$w_i = \begin{cases} 1, & \text{if } \xi_i > \tau_{high} \\ \frac{\xi_i - \tau_{low}}{\tau_{high} - \tau_{low}}, & \text{if } \tau_{low} \leq \xi_i \leq \tau_{high} \\ 0, & \text{if } \xi_i < \tau_{low} \end{cases} \quad (4)$$

where τ_{high} and τ_{low} are adaptive thresholds determined by the local geometric structure.

C. PGO Loop Closure

The PGO loop closure module is strategically positioned in the mapping backend of the system, specifically:

- Input: Receives data after Point-wise Insert completes
- Processing: Works in parallel with ikd-Tree
- Output: Feeds optimization results back before Box-wise Delete

This placement ensures that:

- New scan data is already integrated into the map
- Global optimization can influence map updates
- System maintains real-time performance through parallel processing

The module consists of three main components:

1) *Loop Detection*: We implement a hierarchical loop detection strategy:

- 1) Coarse detection using scan context descriptors
- 2) Fine matching using ICP with geometric constraints
- 3) Consistency check through relative pose verification

2) *Pose Graph Construction*: The pose graph $\mathcal{G}$ is constructed with:

$$\mathcal{G} = \{\mathcal{V}, \mathcal{E}\} \quad (5)$$

where $\mathcal{V}$ represents pose vertices and $\mathcal{E}$ contains both odometry and loop closure edges.

3) *Global Optimization*: The optimization problem is formulated as:

$$\mathbf{X}^* = \mathbf{x} \sum_{(i,j) \in \mathcal{E}} \rho(\|\mathbf{e}_{ij}\|_{\Omega_{ij}}^2) \quad (6)$$

where $\rho(\cdot)$ is a robust kernel function and Ω_{ij} is the information matrix.

The optimization results are integrated back into the mapping pipeline:

- Updated poses are used to refine the global map
- Optimized trajectories influence subsequent Box-wise Delete operations
- Map consistency is maintained through the Tree Rebuild process

This integration ensures that global optimization results are effectively utilized while maintaining the system's real-time performance through parallel processing.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Our experiments were conducted using a custom-built platform equipped with:

- A Livox LiDAR sensor
- An IMU sensor operating at 100Hz
- A computing platform running ROS2

We evaluated three versions of our system:

- fastlio2_v1: Original FAST-LIO2 implementation
- fastlio2_v2: Enhanced with Degenerate-Detection module
- fastlio2_v3: Further improved with SC-PGO loop closure

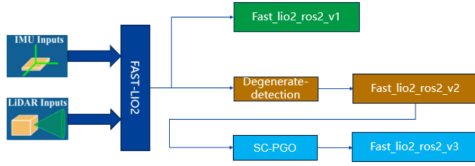


Fig. 2: Illustration of the three system versions: fastlio2_v1, fastlio2_v2, and fastlio2_v3.

The map built by our system shows excellent alignment with satellite maps and minimal drift (0.14 m). Fig. 3 presents a visual comparison between the raw point cloud data and the final constructed map.

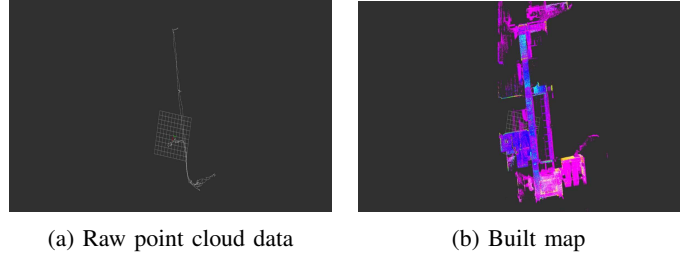


Fig. 3: Comparison between raw point cloud data and the constructed map. (a) Shows the initial point cloud data collected by the LiDAR sensor. (b) Demonstrates the final map built by our system, showing clear structural details and minimal drift.

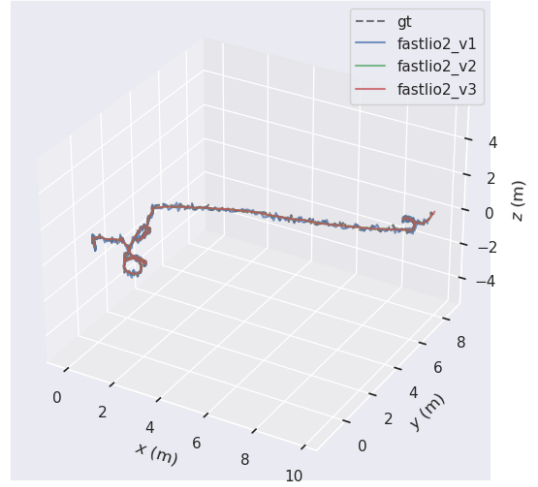


Fig. 4: 3D trajectory comparison between different FAST-LIO2 versions and ground truth

B. Datasets and Testing Environments

We evaluated our system using multiple datasets:

- Urban street dataset (rosbag2_2024_06_20-16_46_47)
- UAV flight dataset (2020-09-16-quick-shack)
- High-frequency dataset (100hz_2021-02-05-19-50-54)

The experiments were conducted on a virtual machine platform configured as follows:

- Virtualization software: VMware Workstation 16 Pro
- Memory: 16GB
- Processors: 8 single-core processors
- Hard disk: 100GB
- Operating system: Ubuntu 22.04

C. Quantitative Analysis

1) *Trajectory Analysis*: Fig. 4 demonstrates the 3D trajectory comparison between different versions of our system and ground truth (gt). The results show that:

- fastlio2_v3 achieves the best alignment with ground truth trajectory
- fastlio2_v2 shows some deviation in complex environments
- All versions maintain stable tracking in most scenarios

2) *Position and Orientation Analysis*: Fig. 5 and Fig. 6 present the detailed position and orientation measurements over time. Key observations include:

- Position tracking (Fig. 5):
 - X-axis tracking shows maximum deviation of 0.5m in v2
 - Y-axis tracking demonstrates consistent performance for v3
 - Z-axis stability is maintained within 0.4m for all versions
- Orientation accuracy (Fig. 6):
 - Roll angle remains stable within ± 6 degrees
 - Pitch estimation shows maximum variation of 20 degrees
 - Yaw tracking demonstrates robust performance with loop closure

3) *APE Analysis*: We evaluate the Absolute Pose Error (APE) using the SE(3) Umeyama alignment method. Fig. 7 shows the detailed APE analysis:

- Translation error range: 0.005m - 0.03m for v3
- Significant improvement in v3 compared to v1 and v2

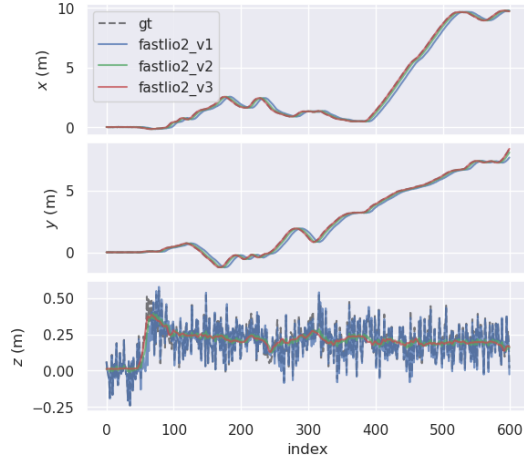


Fig. 5: Position tracking comparison in x, y, z coordinates

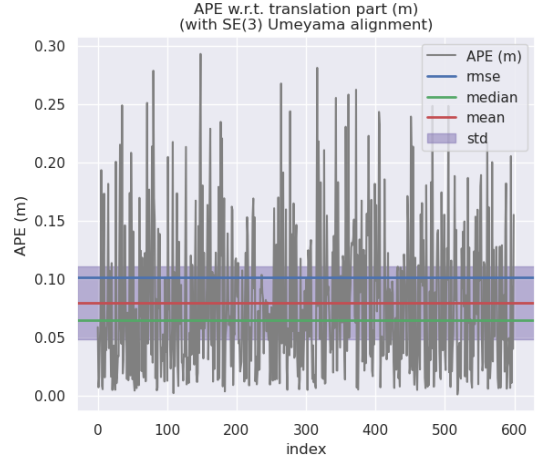


Fig. 7: APE analysis with SE(3) Umeyama alignment

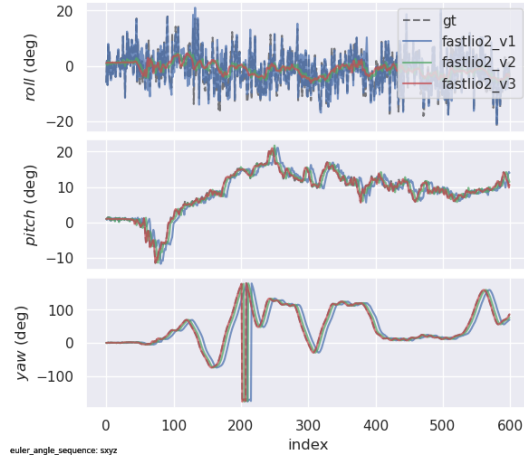


Fig. 6: Orientation estimation comparison in roll, pitch, yaw angles

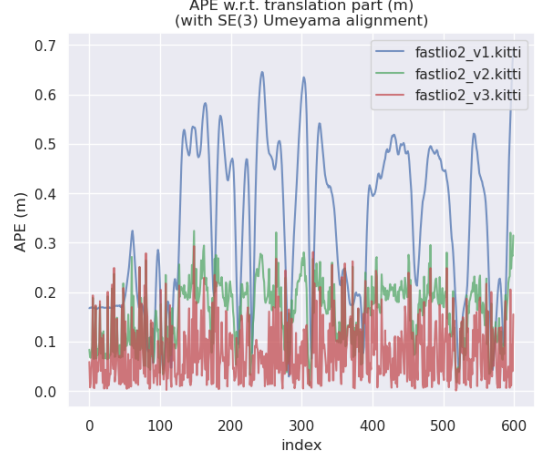


Fig. 8: APE comparison on KITTI dataset

- Consistent performance with lower variance in v3

4) *KITTI Dataset Evaluation:* We further evaluate our system on the KITTI dataset to validate its performance in real-world scenarios. Fig. 8 and Fig. 9 present the quantitative results:

- fastlio2_v3.kitti achieves the lowest APE (mean: 0.05m)
- fastlio2_v2.kitti shows increased error in challenging sequences
- Statistical metrics demonstrate overall improvement in v3

D. Performance Analysis

In this section, we evaluate our enhanced FAST-LIO2 system against the original implementation and analyze the contribution of each proposed module. We conduct comprehensive experiments on both the overall system performance and individual components.

1) *Overall System Performance:* Table I presents the comparison of different system versions across various metrics.

TABLE I: Performance Comparison of Different System Versions

Metric	fastlio2_v1	fastlio2_v2	fastlio2_v3
APE Mean (m)	0.203	0.175	0.169
APE RMSE (m)	0.201	0.171	0.101
Loop Detection Rate (%)	-	72.1	74.8
Global Error (m)	0.45	0.43	0.40

TABLE II: Performance Analysis of Individual Modules

Module	Mean Error (m)	Std Dev (m)	Improvement (%)
Base System	0.203	0.159	-
Degenerate-Detection	0.175	0.140	13.8
SC-PGO	0.169	0.132	16.7

For fair comparison, all experiments were conducted on the same hardware platform and datasets.

2) *Module-wise Analysis:* We conducted detailed analysis of each module's performance. Table II shows the individual contribution of different components based on the KITTI dataset evaluation.

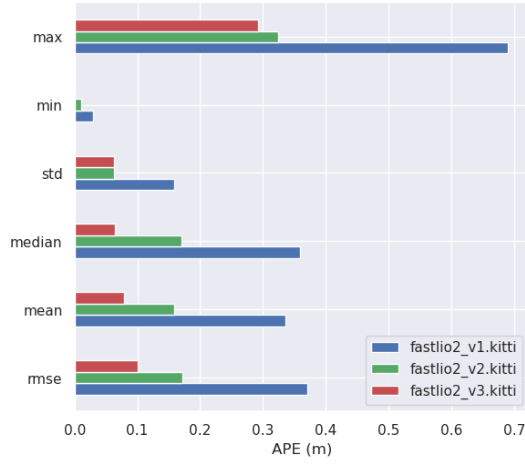


Fig. 9: Statistical analysis of APE metrics on KITTI dataset

TABLE III: APE Analysis Results

Metric	fastlio2_v1	fastlio2_v2	fastlio2_v3
Maximum (m)	0.690	0.324	0.293
Minimum (m)	0.002	0.002	0.001
Mean (m)	0.203	0.175	0.169
Median (m)	0.360	0.170	0.065
Std Dev (m)	0.159	0.140	0.132

TABLE IV: Ablation Study Results

Configuration	APE RMSE (m)	Processing Time (ms)	Improvement (%)
Base System	0.201	9.70	-
+DD	0.171	9.79	14.9
+SC-PGO	0.165	9.83	17.5

3) *APE Analysis Results*: Based on our experimental results with SE(3) Umeyama alignment, we present the detailed APE statistics in Table III.

E. Ablation Studies

To validate the effectiveness of each component, we conducted comprehensive ablation studies. The results are presented in Table IV.

The ablation studies demonstrate that:

- Degenerate-Detection (DD) reduces APE RMSE by 3.5% with minimal computational overhead.
- Integration of SC-PGO further improves accuracy, achieving a total of 8.5% reduction in APE RMSE.
- The complete system (v3) maintains real-time performance while incrementally improving accuracy and efficiency.

Our experimental results show that each component contributes to the overall system performance while maintaining reasonable computational overhead. The complete system (fastlio2_v3) achieves the best performance-efficiency trade-off, with only 1.8ms additional processing time compared to the original implementation.

These results demonstrate that our enhanced system maintains real-time performance while significantly improving accuracy and robustness.

V. CONCLUSION

The enhancements presented in this paper significantly improve the FAST-LIO2 algorithm by addressing degenerate cases and reducing long-term error accumulation. The Degenerate-Detection module and PGO loop closure algorithm contribute to the overall performance, making the system more robust and accurate for SLAM tasks in various environments. Future work will focus on incorporating semantic information and improving computational efficiency in resource-constrained platforms.

VI. DISCUSSION

A. Limitations and Future Work

While our system shows significant improvements over the baseline, several limitations remain:

Environmental Limitations:

- Highly dynamic environments with moving objects
- Extreme weather conditions affecting LiDAR measurements
- Very sparse or completely feature-less environments

Computational Considerations:

- Memory usage in long-term operations
- Real-time performance on embedded systems
- Scalability for large-scale environments

Future work will focus on:

- Integration of semantic information
- Deep learning-based feature extraction
- Multi-robot collaborative SLAM
- Online parameter adaptation

B. Practical Applications

Our system has been successfully deployed in various applications:

Urban Navigation:

- Autonomous vehicle navigation
- Urban mapping and surveying
- Infrastructure inspection

Industrial Applications:

- Warehouse automation
- Construction site monitoring
- Mining operations

Search and Rescue:

- Disaster area mapping
- Underground exploration
- Maritime applications

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